

NETFLIX CONTENT ANALYSIS

End-to-End Data Analytics Project using Excel, MySQL & Tableau

PORTFOLIO PROJECT REPORT

This report documents the complete analytics workflow for a Netflix titles dataset. It covers data cleaning and exploratory analysis in Excel, validation and structured analysis in MySQL, normalized country and genre analysis, and an interactive Tableau dashboard.

Prepared as a Data Analytics Portfolio Project

1. Executive Summary

The project analyzes 8,807 Netflix movies and TV shows to understand content mix, ratings, release patterns, countries, genres, movie duration, and TV-show season distribution. The workflow was completed across Excel, MySQL, and Tableau so that cleaning decisions and major analytical results could be validated across tools.

8,807	6,131	2,676	2021
Total Titles	Movies	TV Shows	Latest Release Year

2. Project Objectives

- Build a clean and analysis-ready Netflix content dataset.
- Measure the distribution of Movies versus TV Shows and identify dominant ratings.
- Analyze how content additions changed over time.
- Identify leading countries and genres using normalized multi-value fields.
- Study movie-duration patterns, TV-show season counts, and release-decade concentration.
- Validate the cleaned dataset and analytical logic in MySQL.
- Build an interactive Tableau dashboard for Movie/TV Show exploration while preserving headline KPIs.

3. Dataset Overview

Attribute	Description
Dataset size	8,807 Netflix titles
Content types	Movies and TV Shows
Release-year range	1925-2021
Core fields	show_id, type, title, director, cast, country, date_added, release_year, rating, duration, listed_in, description
Derived fields	date-added year/month, movie duration minutes, TV seasons, movie duration group
Analysis grain	One row per Netflix title in the main cleaned table

4. Tools & Technologies

Tool	Role in the Project
Microsoft Excel	Data understanding, cleaning, validation, helper columns, PivotTable analysis, charts, business insights, and the Excel dashboard.
MySQL + MySQL Workbench	Database creation, CSV loading, data-type conversion, validation, business analysis, and normalized helper tables for country and genre analysis.
Tableau Desktop	Live MySQL connection, worksheet development, country/genre relationships, KPI cards, interactive filtering, and the final dashboard.

5. Data Cleaning & Preparation

- Verified 8,807 total records and confirmed show_id uniqueness.
- Replaced missing director, cast, and country values with 'Unknown'; retained 10 missing date_added values as genuine blanks.
- Corrected three malformed rating values that were actually duration values and standardized seven missing ratings as 'Not Rated'.
- Validated release years from 1925 through 2021 and confirmed content types were limited to Movie and TV Show.
- Created helper fields for date-added year/month, movie duration in minutes, TV seasons, and movie-duration groups.
- Normalized comma-separated genre values in Excel Power Query for genre-level analysis.
- Preserved distinct rating labels such as NR, UR, and Not Rated rather than combining them.

6. Excel Analysis & Dashboard

Excel was used for the first complete analytical pass. Dedicated analysis sheets covered content type, yearly additions, ratings, countries, genres, movie duration, TV seasons, and release decades. The results were consolidated into a Netflix-themed dashboard.

Analysis	Key Result
Content Type	6,131 Movies (69.62%) and 2,676 TV Shows (30.38%)
Top Rating	TV-MA: 3,207 titles
Top Genre	International Movies: 2,752 titles
Movie Duration	90-119 minutes is the largest group with 3,092 movies
TV Seasons	1-season shows dominate with 1,793 titles
Release Decade	2010-2019 dominates with 5,927 titles

7. MySQL Analysis

The cleaned CSV was loaded into the MySQL database **netflix_content_analysis**. The main table **netflix_titles** contains the cleaned and derived fields used throughout the project. Import validation confirmed 8,807 rows with no import warnings.

- Converted date_added from mixed text formats into a proper DATE field with zero failed conversions.
- Validated duplicate show IDs, missing values, content-type counts, release-year range, ratings, movie-duration values, and TV-season values.
- Used aggregations, GROUP BY, filtering, CASE logic, and other analytical queries to answer business questions.
- Created normalized helper data for multi-country titles so individual countries could be analyzed correctly instead of treating country combinations as single categories.
- Used normalized genre data for Tableau so genre counts reflect individual genres rather than comma-separated combinations.

8. Tableau Data Preparation & Relationships

Tableau connected live to MySQL. The main netflix_titles table was supplemented with normalized helper data. The country helper table was related to the main table using Show ID, allowing titles associated with multiple countries to contribute correctly to country-level analysis. The normalized genre structure supported genre analysis.

Top Country	Normalized Title Count
United States	3,690
India	1,046
United Kingdom	806
Canada	445
France	393
Japan	318
Spain	232
South Korea	231
Germany	226
Mexico	169

9. Tableau Dashboard & Interactivity

The final Tableau dashboard uses a clean white canvas with Netflix red and charcoal accents. Five KPI cards summarize the dataset, while eight analytical visuals provide detailed exploration.

- KPI cards: Total Titles, Movies, TV Shows, Average Movie Duration, and Latest Release Year.
- Visuals: Content Distribution, Content Added Over Time, Top 10 Content Ratings, Top 10 Countries by Content, Movie Duration Distribution, TV Shows by Number of Seasons, Content by Release Decade, and Top 10 Genres.
- The Content Distribution donut acts as a Movie/TV Show dashboard filter.
- Headline KPI cards remain fixed at overall totals while analytical charts respond to the Movie/TV Show selection.
- Redundant field labels, gridlines, and borders were removed for a cleaner presentation.

10. Key Content Insights

- Movies dominate the catalog: 6,131 of 8,807 titles (69.62%), compared with 30.38% TV Shows.
- TV-MA is the most common rating with 3,207 titles, followed by TV-14 with 2,160.
- International Movies is the largest normalized genre with 2,752 titles, followed by Dramas with 2,427 and Comedies with 1,674.
- The United States is the leading normalized country with 3,690 title associations; India is second with 1,046.
- Movies between 90 and 119 minutes form the largest movie-duration group, with 3,092 titles.
- One-season TV shows dominate the TV catalog with 1,793 titles.
- The 2010-2019 release decade contains 5,927 titles, far more than any other decade.
- Content additions accelerated strongly after 2015, peaked around 2019 in the dashboard trend, and were lower by 2021.

11. Analytical Findings & Recommendations

- Use the Movie/TV Show split as a primary segmentation when comparing ratings, countries, and content trends.
- Maintain normalized country and genre structures for future analysis because raw comma-separated fields can distort rankings and counts.
- Separate release year from date added in future trend analysis: release year describes when content was produced, while date added describes when it entered Netflix.
- Future extensions could explore directors, cast members, title age at time of addition, and genre trends by year.

12. Conclusion

This project demonstrates a complete Data Analyst workflow using three complementary tools. Excel was used to clean, validate, explore, and visualize the dataset; MySQL was used to enforce analysis-ready data types, validate records, and perform structured analysis; and Tableau was used to build an interactive, portfolio-ready dashboard. Normalizing multi-value country and genre fields improved analytical accuracy and demonstrated practical data-modeling decisions beyond basic chart creation.

FINAL PROJECT STATUS	STATUS
Excel Cleaning, Analysis & Dashboard	COMPLETE
MySQL Import, Validation & Analysis	COMPLETE
Country & Genre Normalization	COMPLETE
Tableau Dashboard & Interactivity	COMPLETE
Project Documentation	COMPLETE

Project Deliverables

- Cleaned Netflix dataset and Excel analysis workbook.
- Excel dashboard and final dashboard screenshot.
- MySQL database workflow and SQL analysis script.
- Interactive Tableau workbook and final dashboard screenshot.
- Portfolio Project Report for documentation and GitHub presentation.

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Excel | MySQL | Tableau